CASE STUDY

NYC Citi Bike Dashboard (2022)

Python · Pandas · Plotly · Streamlit · Kepler.gl · NOAA API

TOOLS & LIBRARIES

Python · Pandas · NumPy · Plotly · Seaborn · Matplotlib · Streamlit · Kepler.gl

Data source: NYC Citi Bike (2022) — 12 monthly ZIPs, ~1.7M individual trip records

Weather data: NOAA GHCND, station USW00014732 (LaGuardia) — API (TAVG, TMIN, TMAX, PRCP)

Output: Deployed Streamlit dashboard: seasonal trends, temperature, station demand, spatial clusters

github.com/ageelalramadhan/citi-bike-weather-2022

5	1.7M	365	6
Notebooks	Trip Records	Days Merged	Dashboard Pages

PROJECT OVERVIEW

Citi Bike generates millions of trip records each year. This project takes 1.7M individual ride records, merges them with NOAA daily weather data from LaGuardia Airport, and turns the combined dataset into an interactive, deployed dashboard for operational decision-making.

Business question: How should Citi Bike scale bike supply and rebalancing across the year — and which stations, seasons, and environmental factors drive that decision?

DATA PIPELINE

Step	Notebook	Action
`1. Acquire`	citi_bike_weather_2022	Downloaded 12 monthly ZIPs + NOAA API data
`2. Extract`	citi_bike_weather_2022	Unzipped; combined 12 CSVs with generator in pd.concat()
`3. Clean`	citi_bike_weather_2022	Parsed timestamps, filtered 2022, reshaped NOAA long→wide
`4. Aggregate`	citi_bike_weather_2022	Grouped 1.7M rows to daily ride counts
`5. Merge`	citi_bike_weather_2022	Inner join rides + weather → citibike_2022_daily.csv
`6. Visualize`	exercise_2_3 / 2_4	Matplotlib/Seaborn EDA: time series, scatter
`7. Geospatial`	exercise_2_5	Kepler.gl trip origin map
`8. Dashboard`	exercise_2_6	6-page Streamlit app with Plotly
`9. Deploy`	app_part_2.py	Streamlit Cloud deployment

1) SEASONALITY DRIVES DEMAND

Average daily rides climb from ~33,000/day in January to ~115,000/day in August — a 3x+ multiplier.

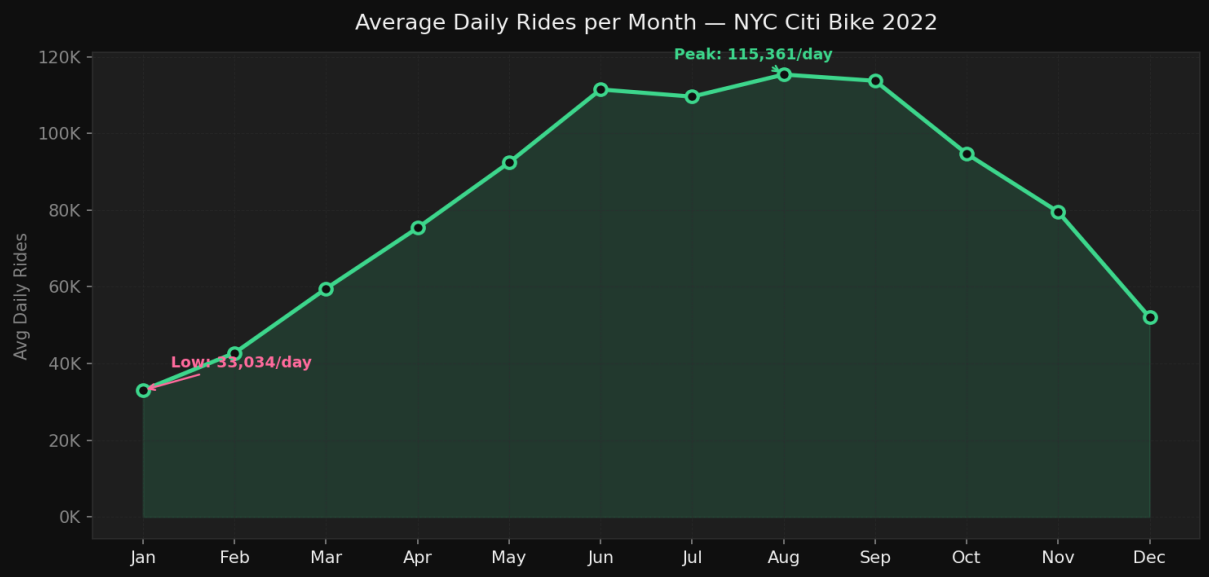

Figure 1. Average daily rides per month, 2022. Peak June–September; winter trough ~50% of summer.

> **Key Insight — Seasonal scaling baseline:**
>
> Winter demand (Nov–Apr) ~50% of summer demand (Jun–Aug). Directly quantifies the seasonal scaling opportunity.

2) TEMPERATURE PREDICTS DAILY RIDERSHIP

Daily ride counts and average temperature track each other closely across the full year.

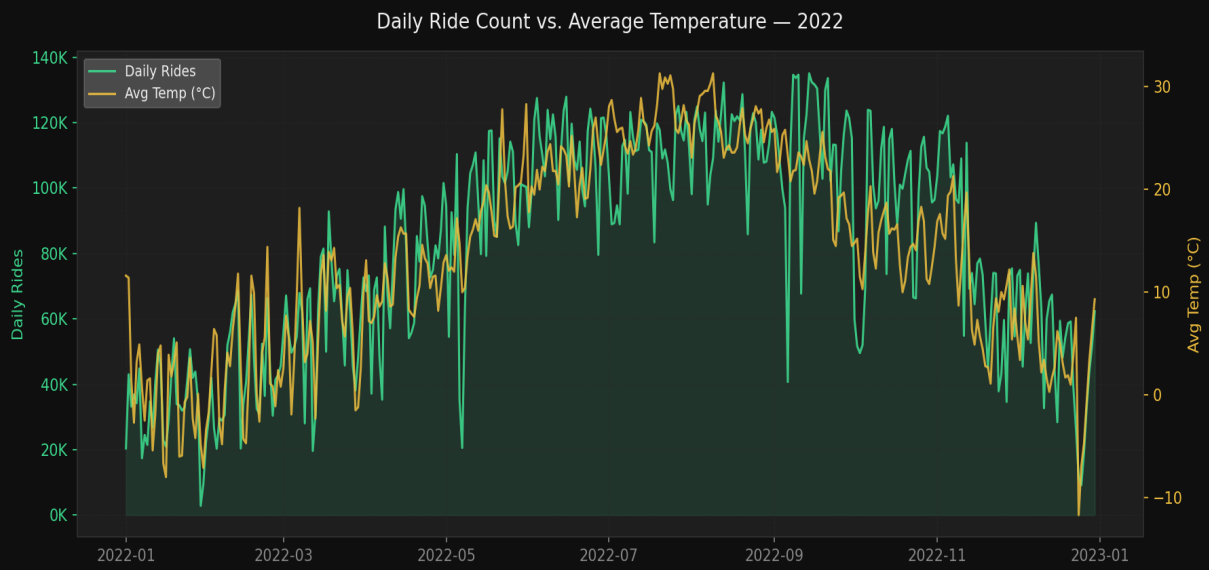

Figure 2. Daily rides (green, left) vs. avg temp °C (yellow, right), dual-axis, 2022.

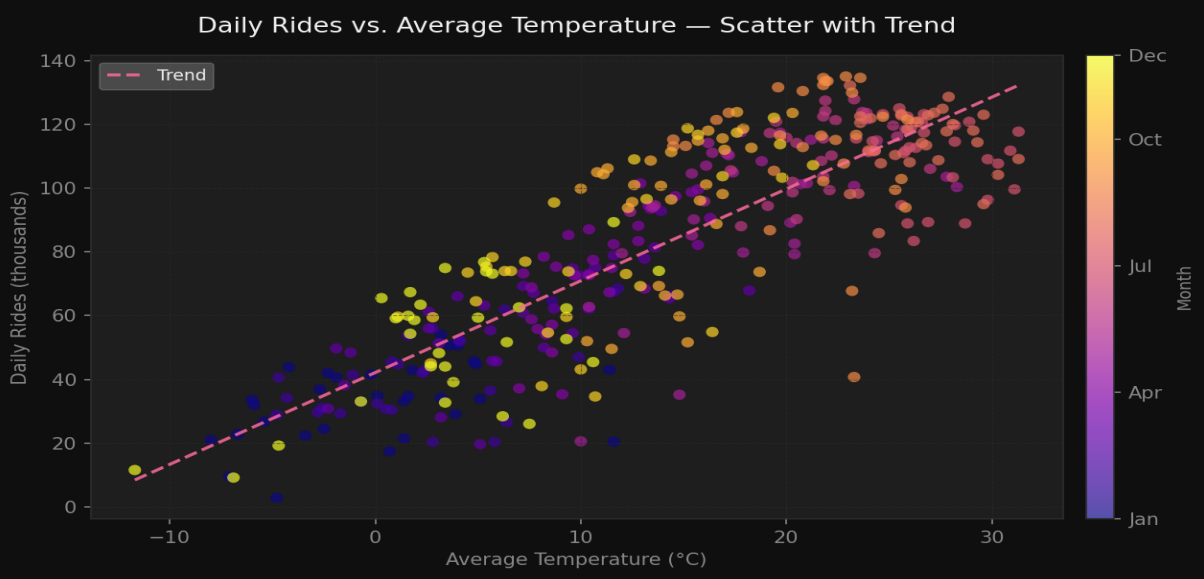

Figure 3. Scatter: daily rides vs. temperature, colour-coded by month. Trend confirms positive relationship.

Key Insight — Weather-aware operations:

NOAA forecasts could be operationalized as a forward-looking demand signal — anticipating spikes before arrival.

3) DEMAND CONCENTRATED IN MANHATTAN

All top-10 start stations in 2022 are in central Manhattan. W 21 St & 6 Ave: 128,436 trip starts.

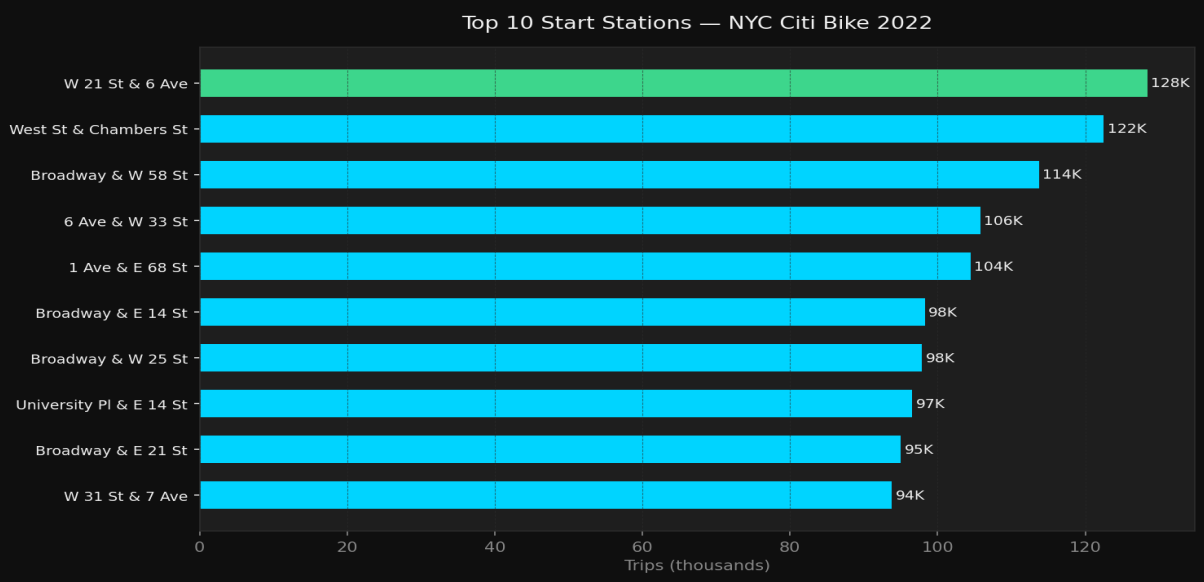

Figure 4. Top 10 start stations, 2022. All in central Manhattan. Data from top10_stations_2022.csv.

Key Insight — Priority rebalancing targets:

These 10 stations are the highest-risk points for empty docks. More frequent rebalancing, higher dock capacity, proactive pre-peak restocking.

KEY FINDINGS

Finding 1 — Winter demand is ~50% of summer demand

August ~115K/day vs. January ~33K/day. 3x+ spread — direct quantified baseline for seasonal scaling.

Finding 2 — Temperature is a reliable demand predictor

Daily rides track temperature closely. NOAA forecasts could be operationalized as a forward-looking demand signal.

Finding 3 — All top-10 stations are in central Manhattan

W 21 St & 6 Ave: 128,436 trip starts. High priority for increased dock capacity and rebalancing frequency.

Finding 4 — Spatial clusters confirm hotspots beyond station rankings

Kepler.gl reveals dense clusters along Hudson waterfront and Brooklyn — movement corridors invisible in station aggregates.

Finding 5 — Generator-based loading required at 1.7M row scale

Generator expression inside pd.concat() streams files sequentially — essential for large-scale local data work.

Live Dashboard: https://citi-bike-weather-2022-f2zr3j3vwcsduf7c5vd84m.streamlit.app/

GitHub: https://github.com/ageelalramadhan/citi-bike-weather-2022

Portfolio: https://ageelalramadhan.github.io